INORGANIC CUMULATIVE EXAM

March 2, 1996

INORGANIC ISOMERIZATION

Answer any <u>six</u> of the following eight questions. Each question is of equal value.

1. Describe or sketch five possible isomers for the carborane $B_7C_2H_9$.

2. Two geometries are common for five-coordinate complexes. What are they? (Give names or sketches). Assume you were able to determine the number of isomers which exist for a complex Ma_3b_2. Show how knowing the number of isomers possible would, or would not, reveal which geometry was being used.

3. The ligand SCN^- is famous for its use in the research of "linkage isomerization". Define "linkage isomerization" and give describe the bonding expected for this ion in complexes of Co(III) and of Pt(II).

4. Sketch or describe two isomers possible for $Co(NH_2CH_2CH_2NH_2)_3^{+3}$, and tell how you could determine if a sample of this complex was all one isomer or was a equal mixture of the two isomers.

5. Describe, name, or sketch the two common geometries found for structural isomerization in complexes with coordination number seven (ML_7).

6. Sketch all of the different isomers possible for the complex Ma_4bc if the complex is (a) octahedral (b) trigonal prismatic.

7. Construct an "all inorganic" (no carbon atoms) complex (of any coordination number) which would be chiral. Use a real metal atom or ion, and real ligands, and give the over-all charge of the complex.

8. Would you expect the same number of isomers to be found for $PtCl_2Br_2^{+2}$ and $SnCl_2Br_2$? Why or why not?